

1 Recurrences

1.1 Fibonacci numbers

Definition 1.1. The Fibonacci numbers are $F_0 = 0$, $F_1 = 1$ and

$$F_{n+2} = F_{n+1} + F_n \quad (n \geq 0). \quad (1.1)$$

Main Theorem. For every $n \geq 1$, $\gcd(F_n, F_{n+1}) = 1$.

Lemma 1.1.1. For every $n \geq 1$,

$$F_{n+1}F_{n-1} - F_n^2 = (-1)^n. \quad (1.2)$$

Proof. We induct on n . For $n = 1$ the left side is $F_2F_0 - F_1^2 = -1$. Assume (1.2) for n . Using (1.1) twice,

$$\begin{aligned} F_{n+2}F_n - F_{n+1}^2 &= (F_{n+1} + F_n)F_n - F_{n+1}(F_n + F_{n-1}) \\ &= F_n^2 - F_{n+1}F_{n-1} \\ &= -(-1)^n = (-1)^{n+1}. \end{aligned} \quad \square$$

Proof of the Main Theorem. A common divisor of F_n and F_{n+1} divides the left side of (1.2), hence divides ± 1 . $\square$

1.2 Strong induction

Theorem 1.2.1. Every positive integer is a sum of distinct, non-consecutive Fibonacci numbers.

Proof. Use strong induction on n . Let F_k be the largest Fibonacci number with $F_k \leq n$. If $F_k = n$ we are done. Otherwise $0 < n - F_k < F_{k-1}$, and by the induction hypothesis $n - F_k$ has such a representation, which cannot use F_{k-1} . $\square$

Fact 1. The ratio F_{n+1}/F_n tends to the golden ratio $\varphi = (1 + \sqrt{5})/2$.

2 Binomial coefficients

Definition 2.1. For integers $0 \leq k \leq n$, let $\binom{n}{k} = \frac{n!}{k!(n-k)!}$.

Theorem 2.0.1 (Pascal's rule). For $1 \leq k \leq n$,

$$\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}. \quad (2.1)$$

Proof. Count k -subsets of $\{1, \dots, n+1\}$ according to whether they contain the element $n+1$. $\square$

Fact 2. Summing Pascal's rule gives $\sum_{k=0}^n \binom{n}{k} = 2^n$.